Smart Life: An ESP32-Based IoT Platform for Integrated Home Automation

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Abstract

Controlling home appliances using multiple independent devices is complicated, and such conditions rarely provide the inhabitants with complete knowledge about the state of the environment or allow timely response to dangerous situations. Smart Life is a prototype developed using the ESP32 DevKit platform that consolidates several methods of controlling connected devices into a single dashboard. The dashboard is hosted on the local Wi-Fi server and allows interacting with relays controlled by a four-channel relay board. The prototype also features infrared emitters programmed to operate televisions and air conditioners. Three sensors are used to monitor the environment: PIR motion sensor, photoresistor, and gas sensor MQ-2. The gas sensor output voltage is used to trigger relays, blink an onboard LED, and send a signal to the dashboard, notifying the user of the gas level threshold being exceeded. The additional functions include learning and re-transmitting infrared signals from consumer electronics remote controllers, user authentication using CAPTCHA, account recovery mechanism based on one-time passwords, and text-to-speech command recognition. Since the dashboard is hosted on the local server, no subscription to external services is necessary to use the prototype. In general, the project demonstrates that microcontroller-based solutions, when combined with appropriate sensors and remote communication methods, can be used to design complete home automation systems. More practical experiments are necessary to evaluate how the prototype will behave under different environmental conditions and ensure that all components operate reliably and securely.

Keywords: IoT; ESP32; home appliances; web dashboard; PIR sensor; LDR; MQ-2; infrared control; embedded systems; wireless communication

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